S. 1.5

3 $f(x) = x + 1/x, g(x) = x - 1/x$

$$\left.\begin{array}{l} D_f = \mathbb{R}^* = \mathbb{R} - \{0\} \\ D_g = \mathbb{R}^* = \mathbb{R} - \{0\} \end{array}\right\} D_{f+g} = D_f \cap D_g = \mathbb{R}^*$$

SOMA:

$$f+g : \mathbb{R}^* \longrightarrow \mathbb{R}$$
$$x \longmapsto (f+g)(x) \overset{DEF}{=} f(x)+g(x)$$

$$(f+g)(x) = \left(x + \cancel{\tfrac{1}{x}}\right) + \left(x - \cancel{\tfrac{1}{x}}\right) = 2x \quad \therefore \quad \boxed{f+g\,(x) = 2x}$$

DIFERENÇA:

$$(f-g) : \mathbb{R}^* \longrightarrow \mathbb{R}$$
$$x \longmapsto (f-g)(x) \overset{DEF}{=} f(x) - g(x)$$

$$(f-g)(x) = f(x) - g(x)$$
$$= \left(x + \tfrac{1}{x}\right) - \left(x - \tfrac{1}{x}\right)$$
$$= \cancel{x} + \tfrac{1}{x} \cancel{-x} + \tfrac{1}{x} = \tfrac{2}{x} \quad \therefore \quad \boxed{(f-g)(x) = \tfrac{2}{x}}$$

PRODUTO:

$$f.g : \mathbb{R}^* \longrightarrow \mathbb{R}$$
$$x \longmapsto f.g(x) \overset{DEF}{=} f(x).g(x)$$

$$f.g\,(x) = f(x).g(x) = \left(x + \tfrac{1}{x}\right)\left(x - \tfrac{1}{x}\right) = x^2 \cancel{-1} \cancel{+1} - \tfrac{1}{x^2} = x^2 - \tfrac{1}{x^2}$$

QUOCIENTE:

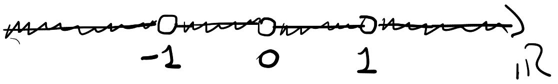
$$\frac{f}{g}(x) \overset{DEF}{=} \frac{f(x)}{g(x)}$$

$g(x) \neq 0 \longleftrightarrow g(x)=0 \iff x-\frac{1}{x}=0$
$\iff x=\frac{1}{x}$
$\iff x^2=1$
$\iff x=\pm 1$

CUIDADO P/ ALÉM DE $D_f \cap D_g$

$\nexists\ \frac{f}{g}(\pm 1)$

$$D_{f/g} \overset{DEF}{=} D_f \cap D_g - \{x;\ g(x)=0\}$$
$$= \mathbb{R}^* - \{-1,1\}$$



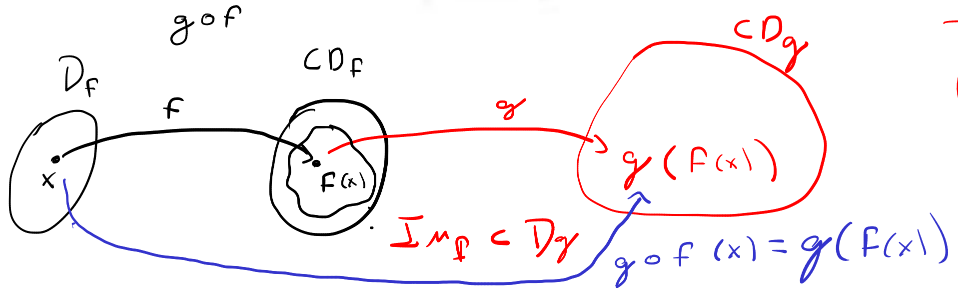
$$\frac{f}{g}(x) = \frac{f(x)}{g(x)} = \frac{x+1/x}{x-1/x} \cdot \overbrace{\frac{x+1/x}{x+1/x}}^{1} = \frac{(x+1/x)^2}{x^2+\cancel{1}-\cancel{1}-1/x^2}$$

$$= \frac{(x+1/x)^2}{x^2-1/x^2} = \frac{x^2+2+1/x^2}{x^2-1/x^2} \,//$$

$$\boxed{\frac{f}{g}(x) = \frac{x^2+1/x^2+2}{x^2-1/x^2}}$$

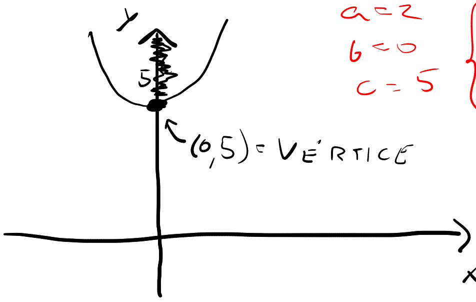
Nos Exercícios 7 a 20 determine $(f \circ g)(x)$ e $(g \circ f)(x)$.

7 $f(x) = 2x^2 + 5,\ g(x) = 4 - 7x$



CUIDADOS DA COMPOSIÇÃO DE FUNÇÕES

$$f : \mathbb{R} \longrightarrow \mathbb{R}$$
$$x \longmapsto 2x^2+5$$



$a = 2$
$b = 0$
$c = 5$

$$\begin{cases} 2x^2+5=0 \\ x^2 = -5/2 \end{cases}$$

$\nexists$ raízes

$$x_v = -\frac{b}{2a} \Rightarrow x_v = \frac{-0}{2.2} = 0$$

$$y_v = f(x_v) = f(0) = 2.(0)^2+5 = 5$$

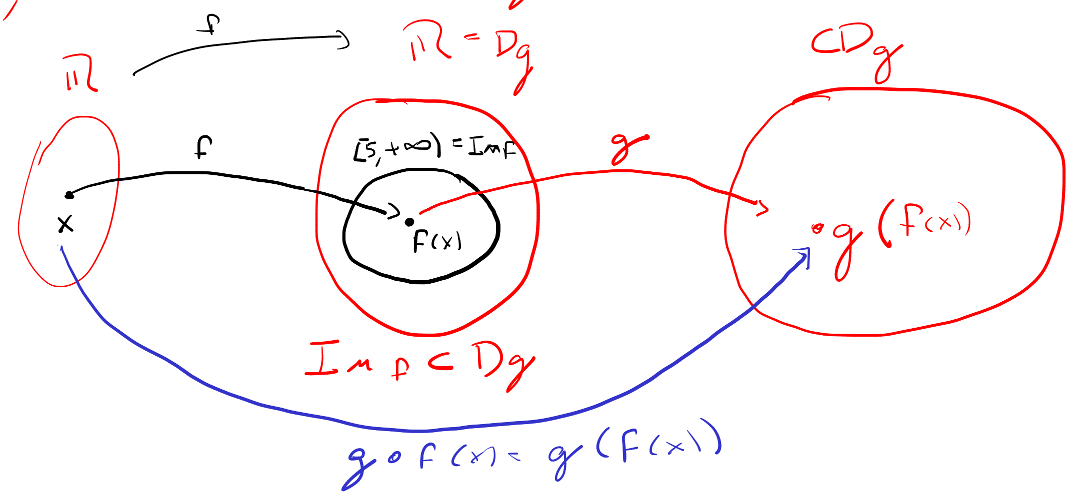
ou

$$= -\frac{\Delta}{4a} = \frac{-(b^2-4ac)}{4a} = \frac{-(0-4.2.5)}{4.2} = \frac{\cancel{4.2}.5}{\cancel{4.2}} = 5$$

Ou seja, $Im_f = [5, +\infty)$ ⊛

ESTUDO DE D_g: $g(x) = 4-7x \therefore D_g = \mathbb{R}$ ⊛

Logo, TEMOS $Im_f \subset D_g$, então $\exists\, g \circ f$.



$$g \circ f : D_f = \mathbb{R} \longrightarrow CD_g$$
$$x \longmapsto g \circ f(x) \overset{DEF}{=} g(f(x))$$

$$g \circ f(x) = g(f(x)) = 4 - 7(f(x))$$

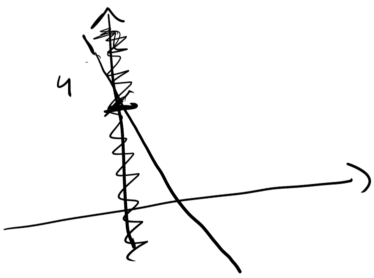
$$g \circ f(x) = 4 - 7\,(f(x))$$
$$= 4 - 7\,.\,(2x^2 + 5)$$
$$= 4 - 14x^2 - 35$$
$$= -14x^2 - 31 \quad \therefore \quad \boxed{g \circ f(x) = -14x^2 - 31}$$

Estudo de $f \circ g(x) \overset{DEF}{=} f(\underbrace{g(x)}_{\hookrightarrow\, Img_g \subset D_f \;\circledast})$

Img: $g(x) = 4 - 7x$

$D_g : \mathbb{R}$

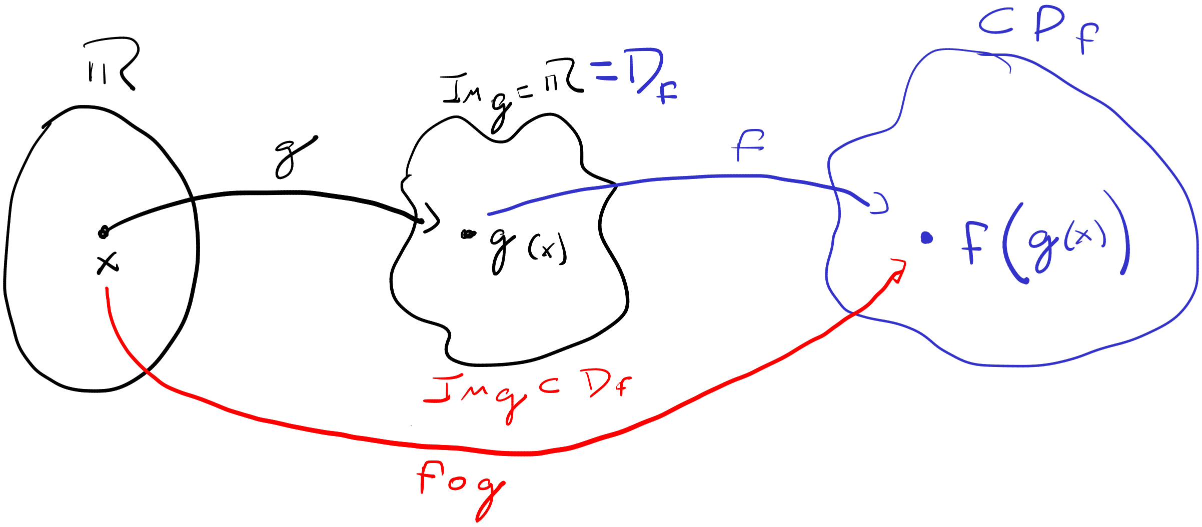
$Img : \mathbb{R}$ $\circledast$



Estudo de D_f:

$f(x) = 2x^2 + 5 \Rightarrow D_f = \mathbb{R}$ $\circledast$

Logo, $Img_g \subset D_f$



$$f \circ g : D_g = \mathbb{R} \longrightarrow CD_f = \mathbb{R}$$

$$x \longmapsto f \circ g(x) \stackrel{DEF}{=} f(g(x))$$

$$
\begin{aligned}
f(g(x)) &= 2(g(x))^2 + 5 \\
&= 2(4-7x)^2 + 5 \\
&= 2(16 - 56x + 49x^2) + 5 \\
&= 32 - 112x + 98x^2 + 5 \\
&= 98x^2 - 112x + 37
\end{aligned}
$$

Obs: $f \circ g \neq g \circ f$